Appendix 4. Included and excluded reports

Deep Learning for Inter-Patient ECG Arrhythmia Classification on the MIT-BIH Database: A Systematic Review

Table A1. Detailed results and transparency items of the 31 included studies

Study [ref]	Type	Classes	S Se/PPV	V Se/PPV	F Se/PPV	Acc	Macro-F1	Confusion matrix	External test	Code	Explainability
CNN-based models											
Xiao H 2022 [20]	C	4	95.54/96.80	90.43/61.32	84.79/95.36	95.38*	89.11* (4 cl.)	Yes	None	No	None
Xie 2023 [21]	J	3	82.73/83.19	94.63/94.22	NR/NR	98.68	92.02 (3 cl.)	Yes	None	No	None
Farag 2023 [22]	J	3	81.60/82.68	95.00/95.63	NR/NR	98.18	92.17 (3 cl.)	No	INCART	No	Matched-filter kernels
Zhou F 2025 [23]	C	4	88.9/65.7	89.6/91.8	0*/NR	96.10	66.04* (4 cl.)	Yes	None	No	t-SNE
Zhang J 2021 [24]	J	5	78.8/90.8	92.5/94.3	NR/NR	94.7	NR	Yes	None	No	t-SNE
Merdjanovska 2021 [25]	C	3	53.52/30.98	82.47/78.44	NR/NR	94.28	71.60 (3 cl.)	No	INCART, SVDB, EST-T	No	None
Berrahou 2026 [26]	J	3	79.02/95.02	96.33/96.79	NR/NR	98.73	94.08* (3 cl.)	No	INCART (retrained on DS1+DS2)	No	None
Zubair 2024 [27]	J	3	96.30/95.31	97.30/97.88	NR/NR	96.19	96.80* (3 cl.)	No	None	No	Attention maps
Li Z 2025 [28]	C	3	72.1/90.3	95.1/94.5	NR/NR	96.3	91.09* (3 cl.)	No	None	No	t-SNE
Cai 2021 [29]	J	5	10.34/28.88	90.93/81.09	0.26/5.00	91.5	49.29* (4 cl.) [a]	Yes	None	No	None
Wang T 2020 [30]	J	4	81.37/68.34	93.72/91.12	5.41/1.06	92.53	66.09 (4 cl.)	Yes	None	Yes	None
Xu P 2022 [31]	J	4	83.06/82.48	93.46/90.04	37.89*/99.32*	98.5	82.02* (4 cl.)	Yes	None	No	Attention maps
CNN-RNN hybrids											
Chen A 2020 [32]	J	4	63.90/76.10	88.84/96.78	47.68/54.25	96.77	77.84 (4 cl.)	Yes	None	No	None
Tao L 2021 [33]	J	4	1.42*/9.09*	84.02*/65.53*	8.64*/1.04*	85.65	42.53* (4 cl.)	Yes	None	No	None
Ran 2024 [34]	J	3	82.88/36.48	95.62/68.91	NR/NR	91.22	75.28* (3 cl.)	Yes	None	No	None
Wang Y 2023 [35]	J	4	73.92/75.57	95.75/85.00	79.64/35.97	95.48	77.99* (4 cl.)	Yes	None	No	t-SNE
Midani 2024 [36]	C	5	92.97/96.01	99.97/99.26	97.42/95.70	99.81	78.08* (5 cl.)	No	None	No	None
Essa 2021 [37]	J	4	65.51/66.19	93.91/94.55	19.33/41.67	95.81	71.06 (4 cl.)	Yes	None	No	t-SNE
Lyu 2023 [38]	J	5	99.67/99.56	100/99.91	100/97.17	99.98	79.62* (5 cl.)	Appendix only	None	No	None
Attention-based models											
Berrahou 2025 [39]	J	3	78.91/94.58	96.68/97.22	NR/NR	98.75	94.12* (3 cl.)	No	INCART (retrained on DS1+DS2)	No	None
Tao Y 2026 [40]	J	4	75.5/91.2	93.6/93.8	52.8/46.6	96.8	81.1 (4 cl.)	Partial	CPSC2018 (rhythm task, not comparable)	No	Attention maps
Other architectures											
Siouda 2021 [41]	J	3	23.98/47.70	86.53/81.65	NR/NR	94.69	71.12* (3 cl.)	No	None	No	None
Zhou C 2023 [42]	J	4	97.66/96.14	99.97/99.07	97.94/98.70	99.88	98.64* (4 cl.)	No	None	Partial	None
Zahid 2023 [43]	J	3	83.37/83.51	91.39/97.38	NR/NR	98.19	92.27* (3 cl.)	Yes	None	Partial	None
Qiu 2021 [44]	J	4	26.90/78.18	92.39/95.74	NR/NR	95.68	NR	No	None	Yes	Contribution heatmaps
Han 2021 [45]	C	4	90.4/57.7	93.3/73.1	21.4/5.0	91.26*	63.91* (4 cl.)	Yes	None	No	Attention weights
Zou 2022 [46]	J	5	78.44*/75.84*	88.45*/94.14*	1.29*/31.25*	96.58*	67.25* (4 cl.) [a]	Yes	None	No	Interpretable context feature
Models with test-patient adaptation											
Chen M 2020 [47]	C	4	78.59/88.67	95.77/93.89	43.81/8.83	94.35	72.5* (4 cl.)	No	None	No	t-SNE
Chang 2026 [48]	J	3	85.03*/93.87*	94.60*/97.04*	NR/NR	98.7	94.78* (3 cl.)	Yes	None	Yes	PCA of clusters
Wang G 2021 [49]	J	4	76.52/90.19	94.02/92.29	57.86/74.10	97.59	84.92* (4 cl.)	No	INCART, SVDB (with adaptation)	Yes	t-SNE
Niu 2020 [50]	J	4	76.6/73.2	85.2/57.8	38.4/44.9	92.3	70.20* (4 cl.)	Yes	None	No	None

Values in %, DS2 test set. *, value computed by the reviewers from the published confusion matrix. NR, not reported or not evaluated (for example, F class in three-class schemes). SVDB, MIT-BIH Supraventricular Arrhythmia Database. EST-T, European ST-T Database. [a] Macro-F1 computed over four classes (N, S, V, F) because the Q class could not be evaluated from the published confusion matrix. Ablation results are given in Appendix 5.

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Table A2. Reports excluded at full-text assessment, with reasons

Authors	Year	Title	Reason for exclusion
Identified through database searching			
Hu D, Li X	2025	Arrhythmia detection algorithm based on domain knowledge and spatio-temporal network model (ICETCI 2025, conference)	F1: custom 20/20 record split, not DS1/DS2
Karmel A, Das J, Rajesh K, Kanimozhi G	2026	An interpretable multi-scale hybrid attention network (MSHAN) for explainable ECG arrhythmia classification (Frontiers in Artificial Intelligence 9:1867173)	F5: non-AAMI scheme (N, L, R, A, V)
Hao S, Xu H, Ji H, Wang Z, Zhao L, Ji Z, Ganchev I	2023	G2-ResNeXt: A Novel Model for ECG Signal Classification (IEEE Access 11)	F5: classifies 3-s segments, not beats
Wang L-H, Yu Y-T, Liu W, et al.	2022	Three-Heartbeat Multilead ECG Recognition Method for Arrhythmia Classification (IEEE Access 10)	F5: classifies 3-beat segments, not beats
Xu SS, Mak M-W, Chang C	2022	Inter-patient ECG classification with i-vector based unsupervised patient adaptation (Expert Systems with Applications 210:118410)	F1: patient-specific protocol (labelled test-record minutes used in training)
Meng Y, Lin L, Qin Z, Qu Y, Qin Y, Li Y	2022	Biosignal Classification Based on Multi-Feature Multi-Dimensional WaveNet-LSTM Models (Journal of Communications 17(5))	F3: only overall accuracy, S/V PPV not derivable
Zhou F, Wang J	2024	Heartbeat classification method combining multi-branch convolutional neural networks and transformer (iScience)	F1: labelled first 5 min of DS2 records added to training
Qin J, Gao F, Wang Z, Liu L, Ji C	2022	Arrhythmia Detection Based on WGAN-GP and SE-ResNet1D (Electronics 11:3427)	F1: custom record split, not DS1/DS2
Arbateni K	2024	Attentional Seq2Seq Encoder-Decoder and Next Generation Reservoir Computing for Heartbeat Classification (IEEE AFROS 2024)	F1: DS1 to DS2 not demonstrable (train/test roles inverted)
Liu M, Shao N, Zheng C, Wang J	2021	Real Time Arrhythmia Monitoring and Classification Based on Edge Computing and DNN (Wireless Communications and Mobile Computing 2021:5563338)	F5: classifies 5-s segments, not beats
Jiang F, Chen J, Gao Y, Sun C	2022	Research on ECG classification based on transfer learning (J China Univ Posts Telecommun 29(6):83-96)	F5: DS2 subsampled and split 80/20 (modified test set)
Chen S, Wang H, Zhang H, Peng C, Li Y, Wang B	2024	A novel method of swin transformer with time-frequency characteristics for ECG-based arrhythmia detection	F1: 10-fold cross-validation grouped by subject
Feng T, Li Q, Zhang Y, Liao Y, Lu D, Wang L, Zhao J, Jiang L, Ni H, Liu H, Deng J	2026	PathFusion-Net: A Rough Path Theory-Based Deep Learning Model for ECG Arrhythmia Classification (IEEE JBHI 30(5):4423-4436, online November 2025)	F1: record 232 moved from DS2 to DS1
Wu X, Rezki Z, Balusu B, Zhou J, Bai J	2026	One model, dual tasks: a novel distributionally adaptive learning framework for ECG classification and generation addressing intra- and inter-patient variability (Expert Syst Appl 299:130096)	F5: DS2 test set completed with synthetic beats
Ben Hnia H, Hmidi A, Khalfallah S, Bouallegue K	2026	ECG-MSEE-Net: A New Model for ECG Heartbeat Classification With Adaptive Spectral Weighting and Its Implementation on an Embedded System (IEEE Access 14)	F1: evaluation protocol not described
Li Y, Qian R, Li K	2022	Inter-patient arrhythmia classification with improved deep residual convolutional neural network (Comput Methods Programs Biomed 214:106582)	F5: classifies 5-s segments, not beats
Siouda R, Nemissi M, Seridi H	2022	A random deep neural system for heartbeat classification (Evolving Systems)	F5: single hidden layer (not deep learning)
Bermeo Cuenca DF, Rodríguez Serrezuela R, Rivera Gómez AE	2025	Arrhythmia Classification from ECG Signals Using LSTM Neural Networks (IEEE AmITIC 2025)	F1: patient-wise GroupKFold cross-validation, not DS1 to DS2
Sengottaiyan N, Sathish Kumar S, Rajesh Sharma R, Sungheetha A, Chinnaiyan R, Hamsanandini S	2024	Compact 1D Self-Operational Neural Networks with Feature Injection for Global ECG Classification (ICIPTM 2024, IEEE)	F1: no evaluation protocol described
Haq SU, Bazai SU, Fatima A, Marjan S, Yang J, Por LY, Anjum M, Shahab S, Ku CS	2023	Reseek-Arrhythmia: Empirical Evaluation of ResNet Architecture for Detection of Arrhythmia (Diagnostics 13(18):2867)	F1: random 70/30 beat split (intra-patient)
Wei G, Deng G, Lu X, Li B	2024	A 1.83 μJ High-Robust Cardiac Health Monitoring with Adaptive-Threshold QRS Detector and Hybrid Neural Network Arrhythmia Classifier (IEEE BioCAS 2024)	F3: only overall accuracy reported, S/V metrics not derivable
Qu Y, Zhang N, Meng Y, Qin Z, Lu Q, Liu X	2020	ECG Heartbeat Classification Detection Based on WaveNet-LSTM (IEEE ICFST 2020)	F1: test on 4,000 resampled samples, not DS2
Tsai IH, Morshed BI	2025	Signal-Specific and Signal-Independent Features for Real-Time Beat-by-Beat ECG Classification with AI for Cardiac Abnormality Detection (Electronics 14(13):2509)	F1: pooled beats split 80/20

Zhou C, Tian L, Li C, Zhang X	2021	Inter-patient ECG Classification Based on Residual Networks and Active Learning (IEEE CISCE 2021)	F1: active learning adds DS2 beats to training
Jangra M, Dhull SK, Singh KK	2020	ECG arrhythmia classification using modified visual geometry group network (mVGGNet) (J Intell Fuzzy Syst 38(3):3151-3165)	F1: custom 30/14 record split
Obeidat Y, Alqudah AM	2021	A Hybrid Lightweight 1D CNN-LSTM Architecture for Automated ECG Beat-Wise Classification (Traitement du Signal 38(5):1281-1291)	F1: pooled beats split randomly
Shoughi A, Dowlatshahi MB	2021	A practical system based on CNN-BLSTM network for accurate classification of ECG heartbeats of MIT-BIH imbalanced dataset (CSICC 2021, IEEE)	F1: 10-fold cross-validation on pooled beats
Oh S, Lee M	2022	A Shallow Domain Knowledge Injection (SDK-Injection) Method for Improving CNN-Based ECG Pattern Classification (Appl Sci 12(3):1307)	F1: other databases (UEA/UCR, PhysioNet), not DS1/DS2
Chen C, Hua Z, Zhang R, Liu G, Wen W	2020	Automated arrhythmia classification based on a combination network of CNN and LSTM (Biomed Signal Process Control 57:101819)	F1: random 10-s segments, 5-fold cross-validation
Nurmaini S, Darmawahyuni A, Sakti Mukti AN, Rachmatullah MN, Firdaus F, Tutuko B	2020	Deep Learning-Based Stacked Denoising and Autoencoder for ECG Heartbeat Classification (Electronics 9(1):135)	F1: two records only, 90/10 beat split
Meng L, Tan W, Ma J, Wang R, Yin X, Zhang Y	2022	Enhancing dynamic ECG heartbeat classification with lightweight transformer model (Artif Intell Med 124:102236)	F1: CPSC2020 database, intra-patient split
Shaker AM, Tantawi M, Shedeed HA, Tolba MF	2020	Generalization of Convolutional Neural Networks for ECG Classification Using Generative Adversarial Networks (IEEE Access 8, 2020)	F1: "DS1/DS2" = original vs GAN-balanced data
Jiang M, Gu J, Li Y, Wei B, Zhang J, Wang Z, Xia L	2021	HADLN: Hybrid Attention-Based Deep Learning Network for Automated Arrhythmia Classification (Front Physiol 12:683025)	F1: PhysioNet/CinC 2017 rhythm database
Irfan S, Anjum N, Althobaiti T, Alotaibi AA, Siddiqui AB, Ramzan N	2022	Heartbeat Classification and Arrhythmia Detection Using a Multi-Model Deep-Learning Technique (Sensors 22(15):5606)	F1: 80/20 split over pooled beats
Tesfai H, Saleh H, Al-Qutayri M, Mohammad MB, Tekeste T, Khandoker A, Mohammad B	2022	Lightweight Shufflenet Based CNN for Arrhythmia Classification (IEEE Access 10, 2022)	F1: random 70/30 segment split
Prabhakararao E, Dandapat S	2022	Multi-Scale Convolutional Neural Network Ensemble for Multi-Class Arrhythmia Classification (IEEE J Biomed Health Inform 26(8):3802)	F1: PTB-XL and CinC 2017 databases
Li J, Pang SP, Xu F, Ji P, Zhou S, Shu M	2022	Two-dimensional ECG-based cardiac arrhythmia classification using DSE-ResNet (Sci Rep 12:14485)	F1: CPSC2018 12-lead database

F1, not trained on DS1 and tested on DS2, or labelled test-patient beats used in training. F3, S/V sensitivity or PPV not reported or derivable. F5, other criterion.